

Akshay Agrawal

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PROFILE

Full-stack software developer with two years of production experience delivering public-sector and B2B products in Java/Spring Boot and TypeScript/Angular. Core contributor on a statewide identity-management platform and on two AI assistants built with LangChain and RAG. Strong testing habit, hexagonal-architecture mindset, and daily use of Claude Code and GitHub Copilot.

WORK EXPERIENCE

PCT digital GmbH, Kiel, Germany — Full-Stack Software Developer Sept 2024 – present

Identity & access management platform · *state-level public-sector IT provider, education sector, statewide rollout* · May 2026 – present

- Shipped account provisioning for a statewide school platform on Java 25 / Spring Boot 4 with hexagonal architecture.
- Automated account handling for schools across the state, from source-system data to directory accounts.
- Built the secured ingest API, multi-source merge rules, and a transactional outbox over RabbitMQ.
- Delivered entitlement rules, Keycloak-based RBAC, and the Vaadin admin UI.
- Owned the entitlement engine end to end: analysed specification gaps, documented decision options for the client, and planned the follow-up tickets.

Waste-management customer app & portal · *municipal-services provider, white-labelled for regional utilities* · Oct 2024 – Jan 2025, Oct 2025 – May 2026

- Built a customer-support AI chatbot end-to-end: LangChain agent, tool calling, RAG on pgvector.
- Enabled the assistant to drive the bulky-waste order form directly inside the Angular UI.
- Closed penetration-test findings (account takeover, IDOR, SQL injection, XSS) across the Ionic/Capacitor iOS/Android app and the Angular portal.

Science-and-industry matching platform · *regional development agency* · Feb – May 2025

- Delivered catalogue, profile, and multi-step registration flows on Angular, tRPC, and PostgreSQL.
- Hardened authentication and data integrity: JWT expiry, Keycloak SSO settings, ownership checks.

AI office assistant for Microsoft 365 · *Teams bot and LLM backend* · Jun – Sept 2025

- Led the Microsoft Teams bot and co-built the LLM backend for an email and calendar assistant.
- Built intent detection, a draft-and-confirm email flow, and scheduled notifications via Microsoft Graph.

Christian-Albrechts-Universität zu Kiel — Research Assistant Apr – Sept 2023

- Information Systems group: analysed the CS curriculum and modelled course relationships in Neo4j with a Flask explorer.

ZeroBS GmbH, Kiel, Germany — Full-Stack Developer (Working Student) Oct 2021 – July 2022

- Built a Flask/JavaScript CMS that tripled content-team productivity.
- Automated operations with one-click deployments, Terraform on AWS/DigitalOcean, Celery, and RabbitMQ.

EDUCATION

Christian-Albrechts-Universität zu Kiel, Germany 2019 – 2023
Master of Science (M.Sc.) | Major: Computer Science

Xavier Institute of Engineering, Mumbai, India 2015 – 2019
Bachelor of Engineering (B.E.) | Major: Computer Science

THESIS & PROJECTS

- **Master thesis — Non-Neural Twin for Simple Autoencoders** ([GitHub](#)): benchmarked non-neural dimensionality reduction against autoencoders; VQPCA matched reconstruction quality on linear data with far better interpretability. Python, NumPy, Keras.
- **Autonomous driving for Formula Student** (master project): centerline estimation and SVM-based track analysis for trajectory optimisation in ROS.
- **Todo-TDD** ([GitHub](#)): Angular, tRPC, and PostgreSQL todo app built strictly test-first.
- **Digital Lending Product** ([GitHub](#)): Django REST API with React frontend, tested with Pytest and Jest.

SKILLS

- **Core stack:** Java (Spring Boot, Spring Data JPA, Spring Security, Maven) · TypeScript/Node.js (Express, tRPC) · Angular · Vaadin · PostgreSQL · RabbitMQ · REST/OpenAPI
- **AI engineering:** LangChain agents with tool calling · RAG on pgvector and Pinecone · Mistral and OpenAI APIs · LLM evaluation and mass testing · Claude Code, GitHub Copilot
- **Security & identity:** Keycloak · OAuth2/OIDC · JWT · RBAC · Microsoft Graph/MSAL · penetration-test remediation
- **Architecture & quality:** Hexagonal architecture · transactional outbox · TDD · JUnit, Jest, Vitest · Cypress/Cucumber · Testcontainers · Liquibase, Prisma
- **Delivery & cloud:** Docker · GitHub Actions, GitLab CI · Helm charts and local Kubernetes (k3d, Tilt) · OpenTofu/Terraform · Ansible · Open Telekom Cloud · Nx monorepos · Git, Agile/Scrum
- **Mobile:** Ionic, Capacitor (iOS/Android)

LANGUAGES & INTERESTS

English (fluent) · German (B1, actively improving) · Hindi (native) | Photography, swimming, cooking, volleyball